

Heat Transfer in a Helical Coil

April 24, 2026

Abstract

This report investigates the convective heat transfer coefficient (h) of Newtonian fluids (pure water and glycerol-water mixtures at 10%, 15%, 20%, and 30% v/v) flowing through a helically coiled copper tube under an isothermal wall boundary condition. Centrifugal forces in the curved geometry induce secondary flows (Dean vortices), which enhance radial mixing and overall heat transfer efficiency. The system exhibited a severe “thermal pinch,” with exit temperatures equalizing within $1^{\circ}\text{C} - 2.6^{\circ}\text{C}$ of the bath temperature across all flow conditions. The analysis establishes a clear inverse relationship between the fluid mixture viscosity and the experimental convective heat transfer coefficient. Furthermore, the experimental Nusselt numbers were validated against the theoretical model proposed by Dravid et al. (1971) for thermally developing laminar flow. The data showed excellent agreement for pure water and moderate viscosities (10%-20% v/v), mathematically proving that the thermal boundary layer was still developing within the 3.0-meter coil. The 30% glycerol runs were explicitly excluded from this theoretical comparison, as their extreme viscosity resulted in Dean numbers below the model’s absolute validity threshold ($De < 50$).

Contents

1	Nomenclature	3
2	Objective	3
3	Detailed Theory	4
3.1	Heat Transfer in Helical Coils and the Dean Effect	4
3.2	Dimensionless Numbers Definition	4
3.3	Theoretical Benchmark for Developing Flow (Dravid et al.)	5
3.4	Theoretical Baseline for Straight Pipe (Hausen Correlation)	5
3.5	The Log Mean Temperature Difference (LMTD)	5
3.6	Axial Temperature Profile Prediction	6
3.7	Evaluation of Fluid Properties (T_{bulk})	6
3.7.1	Temperature & Mixture Correlations	6
4	Materials & Equipment	8
4.1	The Core Apparatus	8
4.2	The Fluid Delivery System	8
5	Sample Calculations	9
5.1	Sample 1: 10% Glycerol at Low Bath Temperature	9
5.2	Sample 2: 30% Glycerol at High Bath Temperature	11
6	Results & Data Analysis	12
7	Discussion	20
7.1	The Thermal Pinch Effect (Figure 2)	20
7.2	Impact of Viscosity (Figure 3)	20
7.3	Interpretation of the Viscosity-Heat Transfer Relationship (Figures 4 and 5)	21
7.4	Validation Against the Dravid et al. Theoretical Model	21
7.5	Geometric Efficiency Gain over Straight Pipes	21
8	Error Analysis & Discussion	22
8.1	Propagation of Uncertainty (Logarithmic Method)	22
9	Conclusion	24

1 Nomenclature

Symbol	Description	Units
A_i	Inner surface area of the helical coil	m^2
C_p	Specific heat capacity of the fluid	$\text{J}/(\text{kg} \cdot \text{K})$
D_c	Major diameter of the helical coil	m
D_i	Inner diameter of the copper tube	m
De	Dean number	Dimensionless
h_{exp}	Experimental convective heat transfer coefficient	$\text{W}/(\text{m}^2 \cdot \text{K})$
k	Thermal conductivity of the fluid mixture	$\text{W}/(\text{m} \cdot \text{K})$
L	Total uncoiled length of the tube	m
$\dot{m}$	Mass flow rate of the fluid	kg/s
Nu	Nusselt number	Dimensionless
Pr	Prandtl number	Dimensionless
$\dot{Q}$	Heat	W
$\dot{V}$	Volumetric flow rate	m^3/s
Re	Reynolds number	Dimensionless
T_{bath}	Isothermal water bath temperature	$^{\circ}\text{C}$
T_{bulk}	Bulk film temperature for property evaluation	$^{\circ}\text{C}$
T_{in}	Inlet temperature of the fluid	$^{\circ}\text{C}$
T_{out}	Outlet temperature of the fluid	$^{\circ}\text{C}$
ΔT_{lm}	Log Mean Temperature Difference (LMTD)	$^{\circ}\text{C}$
μ	Dynamic viscosity of the fluid	$\text{kg}/(\text{m} \cdot \text{s})$
ρ	Density of the fluid	kg/m^3
w_i	Mass fraction of component i	Dimensionless

2 Objective

To study the convective heat transfer characteristics of Newtonian fluids (pure water and glycerol-water mixtures) in a helical coil heat exchanger under an isothermal wall boundary condition. The specific objectives are:

- To determine the experimental convective heat transfer coefficient (h_{exp}) for varying concentrations of glycerol (0%, 10%, 15%, 20%, and 30% v/v).

- To observe the effect of increasing fluid viscosity on the overall heat transfer rate and quantify its deterioration.
- To validate the experimental data against established theoretical models for developing flow in curved pipes.

3 Detailed Theory

3.1 Heat Transfer in Helical Coils and the Dean Effect

When a fluid flows through a straight pipe, the velocity profile is parabolic, and heat transfer relies heavily on conduction through the fluid layers in the laminar regime. However, in a helically coiled tube, the fluid experiences centrifugal forces due to the curvature. The fluid at the centerline, moving at the highest axial velocity, is subjected to a greater centrifugal force than the slower-moving fluid near the tube walls.

This imbalance forces the cooler, heavier core fluid outward toward the heated wall, while the fluid near the wall is displaced inward. This continuous secondary flow pattern takes the form of two counter-rotating vortices, known as **Dean vortices**. These vortices violently disrupt the thermal boundary layer, promoting rapid radial mixing. Consequently, the convective heat transfer coefficient (h) in a coiled tube is significantly higher than in a straight tube of identical length and diameter operating at the same Reynolds number.

3.2 Dimensionless Numbers Definition

To generalize the heat transfer behavior across different fluids and operating conditions, several dimensionless parameters are utilized:

- **Reynolds Number (Re):** Ratio of inertial forces to viscous forces. Defines the primary flow regime. $Re = \frac{\rho V D_i}{\mu}$
- **Prandtl Number (Pr):** Ratio of momentum diffusivity to thermal diffusivity. $Pr = \frac{c_p \mu}{k}$
- **Nusselt Number (Nu):** Ratio of convective to conductive heat transfer across the boundary layer. $Nu = \frac{h D_i}{k}$
- **Dean Number (De):** A modified Reynolds number that characterizes the magnitude of the secondary flow (Dean vortices) in a curved pipe. It relates the inertial, viscous, and centrifugal forces.

$$De = Re \sqrt{\frac{D_i}{D_c}} \quad (1)$$

3.3 Theoretical Benchmark for Developing Flow (Dravid et al.)

Because highly viscous fluids take a significant distance to achieve a fully developed temperature profile, experimental data in relatively short coils must be compared against developing flow models.

In their seminal research paper, *Effect of Secondary Fluid Motion on Laminar Flow Heat Transfer in Helically Coiled Tubes* (A. N. Dravid, K. A. Smith, E. W. Merrill, and P. L. T. Brian, Massachusetts Institute of Technology, 1971), a robust theoretical correlation was established for thermally developing laminar flow. They demonstrated that the Nusselt number (Nu_c) in a coiled tube under these conditions can be accurately predicted using the Dean number (De).

The Dravid et al. correlation is given by the equation:

$$Nu_c = \left[0.76 + 0.65\sqrt{De} \right] Pr^{0.175} \quad (2)$$

Constraint Boundary: This theoretical model is empirically derived and strictly valid only for Dean numbers in the range $50 < De < 2000$. Experimental data points falling below a Dean number of 50 lack sufficient centrifugal force to sustain the secondary vortices required by this specific model, and therefore cannot be accurately benchmarked against this equation.

3.4 Theoretical Baseline for Straight Pipe (Hausen Correlation)

To demonstrate the immense efficiency gain provided by the Dean vortices, the experimental data from the helical coil is benchmarked against the theoretical heat transfer coefficient of a perfectly straight pipe of identical length and diameter. The widely accepted Hausen correlation for thermally developing laminar flow in a straight pipe is used for this baseline:

$$Nu_{straight} = 3.66 + \frac{0.065 \left(\frac{D_i}{L} \right) RePr}{1 + 0.04 \left[\left(\frac{D_i}{L} \right) RePr \right]^{2/3}} \quad (3)$$

By comparing the physical results of the coil against this theoretical straight-pipe limit, the multiplying power of the centrifugal forces can be directly quantified.

3.5 The Log Mean Temperature Difference (LMTD)

Because the temperature of the fluid changes continuously as it travels through the coil, the driving force for heat transfer is not constant. The appropriate average temperature

difference for an isothermal wall condition is the LMTD (ΔT_{lm}):

$$\dot{Q} = h_{exp} A_i \Delta T_{lm} = h_{exp} A_i \frac{(T_{bath} - T_{in}) - (T_{bath} - T_{out})}{\ln \left(\frac{T_{bath} - T_{in}}{T_{bath} - T_{out}} \right)} \quad (4)$$

By equating this to the sensible heat gained by the fluid ($\dot{Q} = \dot{m} C_p (T_{out} - T_{in})$), the experimental heat transfer coefficient is calculated as:

$$h_{exp} = \left(\frac{\dot{m} \cdot C_p}{A_i} \right) \cdot \ln \left(\frac{T_{bath} - T_{in}}{T_{bath} - T_{out}} \right) \quad (5)$$

3.6 Axial Temperature Profile Prediction

To mathematically visualize the thermal pinch within the coil, the axial temperature profile $T(x)$ at any distance x along the pipe can be predicted using the macroscopic energy balance for a constant-temperature wall:

$$T(x) = T_{bath} - (T_{bath} - T_{in}) \exp \left(-\frac{h \cdot \pi \cdot D_i \cdot x}{\dot{m} \cdot C_p} \right) \quad (6)$$

By substituting the experimental coiled heat transfer coefficient (h_{exp}) and the theoretical straight-pipe coefficient ($h_{straight}$, derived from the Hausen correlation), the immense spatial efficiency of the Dean vortices can be plotted and compared directly against a straight pipe geometry.

3.7 Evaluation of Fluid Properties (T_{bulk})

The properties of glycerol-water mixtures (viscosity, thermal conductivity, density, specific heat) are highly temperature-dependent. Evaluating these properties at the room-temperature inlet (T_{in}) leads to severe inaccuracies.

Because the apparatus exhibits a thermal pinch (where $T_{out} \approx T_{bath}$), the fluid spends the vast majority of its residence time inside the coil at a temperature closely approaching the bath temperature. Therefore, fluid properties are evaluated at the arithmetic **Bulk Film Temperature**:

$$T_{bulk} = \frac{T_{in} + T_{bath}}{2} \quad (7)$$

3.7.1 Temperature & Mixture Correlations

Pure component properties are evaluated dynamically at T_{bulk} using standard empirical regressions. Once pure values at T_{bulk} are established, mixture correlations are applied:

1. Thermal Conductivity (k): Pure component thermal conductivities (in W/(m·K)) are evaluated as a function of temperature (T in K) using correlations from Yaws'

Chemical Properties Handbook (McGraw-Hill, 1999):

For Water (H_2O):

$$k_w = -0.2758 + 4.6120 \times 10^{-3}T - 5.5391 \times 10^{-6}T^2 \quad (8)$$

For Glycerol ($C_3H_8O_3$):

$$\log_{10}(k_g) = -0.3550 - 0.2097 \left[1 - \frac{T}{723.00} \right]^{2/7} \quad (9)$$

The mixture thermal conductivity is then calculated using the **Filippov correlation** for binary liquids:

$$k_{mix} = w_w k_w + w_g k_g - 0.72 w_w w_g (k_w - k_g) \quad (10)$$

Table 1: Thermal Conductivity (k_{mix}) of Glycerol-Water Mixtures at Target Bulk Temperatures

Concentration (% v/v)	k_{mix} at 38.5°C (W/m·K)	k_{mix} at 43.0°C (W/m·K)
0% (Water)	0.6235	0.6286
10%	0.5571	0.5613
15%	0.5277	0.5315
20%	0.5007	0.5040
30%	0.4531	0.4557

2. Dynamic Viscosity (μ_{mix}): The dynamic viscosity of the glycerol-water mixtures was determined using an online calculator hosted by the University of Reading. This tool calculates mixture properties based on the parameterization models developed by Cheng (2008), incorporating subsequent refinements for volume contraction and temperature-dependence established by Volk and Kähler (2018). The specific dynamic viscosity values utilized for both target bulk temperatures are outlined in Table 2.

Table 2: Dynamic Viscosity (μ_{mix}) of Glycerol-Water Mixtures at Target Bulk Temperatures

Concentration (% v/v)	μ_{mix} at 38.5°C (kg/m·s)	μ_{mix} at 43.0°C (kg/m·s)
0% (Water)	0.000670	0.000610
10%	0.00090170	0.00082449
15%	0.0010548	0.00096095
20%	0.0012444	0.00112888
30%	0.0017836	0.0016011

4 Materials & Equipment

4.1 The Core Apparatus

- **Material:** Soft copper tubing (23 SWG), chosen for high thermal conductivity.
- **Dimensions:** Length (L) = 3.0 m. Inner diameter (D_i) = 8.31 mm.
- **Geometry:** Major coil diameter (D_c) = 150 mm. Pitch = 15 mm.
- **Water Bath:** A cuboidal bath with a thermostat and continuous mechanical agitation, providing a perfect isothermal wall boundary condition.

4.2 The Fluid Delivery System

The working fluid is actively pumped to overcome the high pressure drop of the viscous glycerol mixtures and ensure precise residence times.

- **Reservoir & Pump:** Fluid is stored in a red bucket on the floor. A submersible mechanical pump sits at the base.
- **Rotameter:** Transparent tubing carries the fluid to a glass-tube rotameter mounted on a control panel, calibrated to maintain exactly 20 LPH.
- **Thermometry:** Digital thermometers record T_{in} before the bath and T_{out} immediately upon exit.

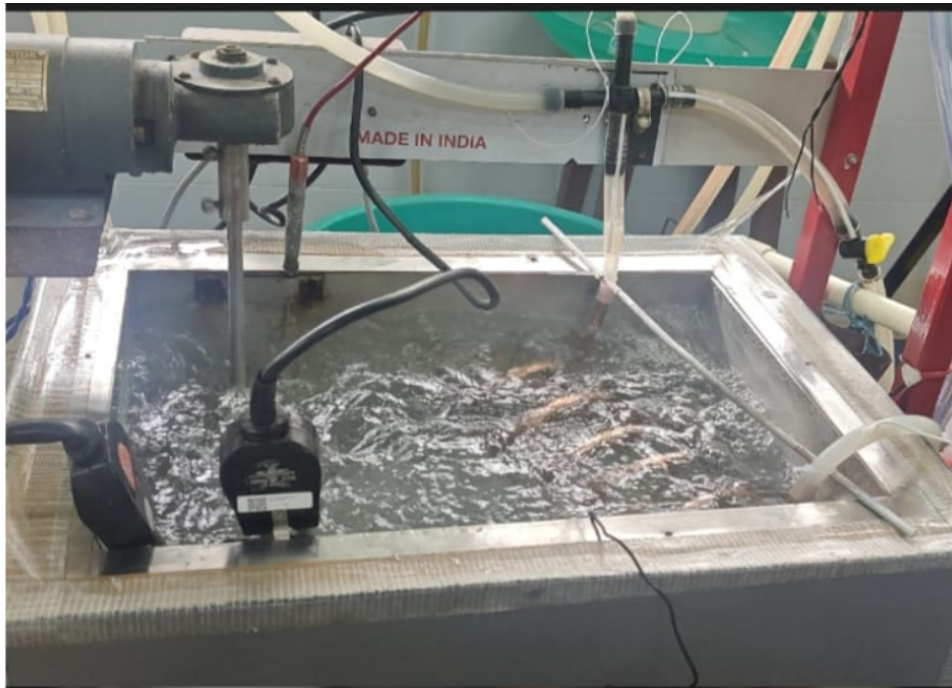


Figure 1: Diagram/Photograph of the Helical Coil Apparatus

5 Sample Calculations

5.1 Sample 1: 10% Glycerol at Low Bath Temperature

Raw Experimental Data (Run 3):

- Concentration (V/V) = 10%
- Volumetric Flow Rate ($\dot{V}$) = 20 LPH = $5.556 \times 10^{-6} \text{ m}^3/\text{s}$
- Temperatures: $T_{in} = 25.9^\circ\text{C}$, $T_{bath} = 51.0^\circ\text{C}$, $T_{out} = 49.8^\circ\text{C}$
- Inner Tube Diameter (D_i) = 0.00831 m
- Coil Diameter (D_c) = 0.150 m

Step 1: Determine Bulk Temperature (T_{bulk})

$$T_{bulk} = \frac{T_{in} + T_{out}}{2} \approx \frac{25.9 + 51.0}{2} = 38.5^\circ\text{C} \quad (11)$$

Step 2: Mixture Properties at T_{bulk}

Based on established property tables and the University of Reading mixture calculator for viscosity at 38.5°C :

- Density (ρ_{mix}) = 1018.6 kg/m^3
- Specific Heat ($C_{p,mix}$) = $3973 \text{ J/(kg} \cdot \text{K)}$
- Thermal Conductivity (k_{mix}) = $0.5571 \text{ W/(m} \cdot \text{K)}$
- Dynamic Viscosity (μ_{mix}) = $0.00090170 \text{ kg/(m} \cdot \text{s)}$

Step 3: Log Mean Temperature Difference (ΔT_{lm})

$$\begin{aligned} \Delta T_{in} &= T_{bath} - T_{in} = 51.0 - 25.9 = 25.1^\circ\text{C} \\ \Delta T_{out} &= T_{bath} - T_{out} = 51.0 - 49.8 = 1.2^\circ\text{C} \\ \Delta T_{lm} &= \frac{25.1 - 1.2}{\ln(25.1/1.2)} = 7.86^\circ\text{C} \end{aligned}$$

Step 4: Experimental Heat Transfer Coefficient (h_{exp})

$$\begin{aligned} \dot{m} &= \rho_{mix} \cdot \dot{V} = 1018.6 \times (5.556 \times 10^{-6}) = 5.67 \times 10^{-3} \text{ kg/s} \\ \dot{Q} &= \dot{m} C_{p,mix} (T_{out} - T_{in}) = (5.67 \times 10^{-3})(3973)(49.8 - 25.9) = 538.4 \text{ W} \\ h_{exp} &= \frac{\dot{Q}}{A_i \cdot \Delta T_{lm}} = 875 \text{ W/(m}^2 \cdot \text{K)} \\ Nu_{exp} &= \frac{h_{exp} D_i}{k_{mix}} = \frac{875 \cdot 0.00831}{0.5571} = 13.05 \end{aligned}$$

Step 5: Dravid et al. Theoretical Validation (h_{th})

$$\begin{aligned} Re &= \frac{4\dot{m}}{\pi D_i \mu_{mix}} = \frac{4(5.67 \times 10^{-3})}{\pi(0.00831)(0.00090170)} = 964 \\ Pr &= \frac{\mu_{mix} C_{p,mix}}{k_{mix}} = \frac{0.00090170 \times 3973}{0.5571} = 6.43 \\ De &= Re \sqrt{\frac{D_i}{D_c}} = 964 \sqrt{\frac{0.00831}{0.150}} = 227 \\ Nu_{th} &= \left[0.76 + 0.65 \sqrt{De} \right] Pr^{0.175} = \left[0.76 + 0.65 \sqrt{227} \right] 6.43^{0.175} = 14.61 \\ h_{th} &= \frac{Nu_{th} k_{mix}}{D_i} = \frac{14.61 \cdot 0.5571}{0.00831} = 980 \text{ W/(m}^2 \cdot \text{K)} \end{aligned}$$

$$\text{Deviation: } \frac{h_{exp} - h_{th}}{h_{th}} \times 100 = \frac{875 - 980}{980} \times 100 \approx -11\%$$

5.2 Sample 2: 30% Glycerol at High Bath Temperature

Raw Experimental Data (Run 10):

- Concentration (V/V) = 30%
- Volumetric Flow Rate ($\dot{V}$) = 20 LPH = $5.556 \times 10^{-6} \text{ m}^3/\text{s}$
- Temperatures: $T_{in} = 25.9^\circ\text{C}$, $T_{bath} = 60.8^\circ\text{C}$, $T_{out} = 58.6^\circ\text{C}$

Step 1: Determine Bulk Temperature (T_{bulk})

$$T_{bulk} = \frac{T_{in} + T_{bath}}{2} = \frac{25.9 + 60.8}{2} = 43.35^\circ\text{C} \approx 43.0^\circ\text{C} \quad (12)$$

Step 2: Mixture Properties at T_{bulk}

Using properties at 43.0°C with updated Univ. of Reading viscosity model:

- Density (ρ_{mix}) = 1068.2 kg/m^3
- Specific Heat ($C_{p,mix}$) = $3591 \text{ J/(kg} \cdot \text{K)}$
- Thermal Conductivity (k_{mix}) = $0.4557 \text{ W/(m} \cdot \text{K)}$
- Dynamic Viscosity (μ_{mix}) = $0.0016011 \text{ kg/(m} \cdot \text{s)}$

Step 3: Log Mean Temperature Difference (ΔT_{lm})

$$\begin{aligned} \Delta T_{in} &= T_{bath} - T_{in} = 60.8 - 25.9 = 34.9^\circ\text{C} \\ \Delta T_{out} &= T_{bath} - T_{out} = 60.8 - 58.6 = 2.2^\circ\text{C} \\ \Delta T_{lm} &= \frac{34.9 - 2.2}{\ln(34.9/2.2)} = 11.83^\circ\text{C} \end{aligned}$$

Step 4: Experimental Heat Transfer Coefficient (h_{exp})

$$\begin{aligned} \dot{m} &= \rho_{mix} \cdot \dot{V} = 1068.2 \times (5.556 \times 10^{-6}) = 5.98 \times 10^{-3} \text{ kg/s} \\ \dot{Q} &= \dot{m} C_{p,mix} (T_{out} - T_{in}) = (5.98 \times 10^{-3})(3591)(58.6 - 25.9) = 702.2 \text{ W} \\ h_{exp} &= \frac{\dot{Q}}{A_i \cdot \Delta T_{lm}} = 758 \text{ W/(m}^2 \cdot \text{K)} \\ Nu_{exp} &= \frac{h_{exp} D_i}{k_{mix}} = \frac{758 \cdot 0.00831}{0.4557} = 13.82 \end{aligned}$$

Step 5: Dravid et al. Theoretical Validation (h_{th})

$$Re = \frac{4\dot{m}}{\pi D_i \mu_{mix}} = \frac{4(5.98 \times 10^{-3})}{\pi(0.00831)(0.0016011)} = 572$$

$$Pr = \frac{\mu_{mix} C_{p,mix}}{k_{mix}} = \frac{0.0016011 \times 3591}{0.4557} = 12.62$$

$$De = Re \sqrt{\frac{D_i}{D_c}} = 572 \sqrt{\frac{0.00831}{0.150}} = 135$$

$$Nu_{th} = \left[0.76 + 0.65 \sqrt{De} \right] Pr^{0.175} = \left[0.76 + 0.65 \sqrt{135} \right] 12.62^{0.175} = 12.95$$

$$h_{th} = \frac{Nu_{th} k_{mix}}{D_i} = \frac{12.95 \cdot 0.4557}{0.00831} = 710 \text{ W}/(\text{m}^2 \cdot \text{K})$$

Deviation: $\frac{h_{exp} - h_{th}}{h_{th}} \times 100 = \frac{758 - 710}{710} \times 100 \approx 7\%$

6 Results & Data Analysis

Table 3: Observation Table: Temperature Readings and LMTD

Run	Conc (% v/v)	T_{bath} (°C)	T_{in} (°C)	T_{out} (°C)	ΔT_{in} (°C)	ΔT_{out} (°C)	ΔT_{lm} (°C)
1	0% (Water)	51.2	25.9	50.6	25.3	0.6	6.60
2	0% (Water)	61.3	25.9	60.4	35.4	0.9	9.40
3	10%	51.0	25.9	49.8	25.1	1.2	7.86
4	10%	60.0	25.9	58.2	34.1	1.8	10.98
5	15%	51.1	25.9	48.9	25.2	2.2	9.43
6	15%	61.2	25.9	59.5	35.3	1.7	11.08
7	20%	53.0	25.9	51.5	27.1	1.5	8.85
8	20%	60.5	25.9	57.9	34.6	2.6	12.36
9	30%	51.3	25.9	49.7	25.4	1.6	8.61
10	30%	60.8	25.9	58.6	34.9	2.2	11.83

Table 4: Fluid Properties at Bulk Temperatures

Concentration	ρ at 38.5°C (kg/m ³)	C_p at 38.5°C (J/kg·K)	ρ at 43.0°C (kg/m ³)	C_p at 43.0°C (J/kg·K)
0% (Water)	992.8	4179	991.1	4179
10%	1021.1	3973	1019.2	3973
15%	1035.3	3874	1033.4	3874
20%	1049.5	3777	1047.5	3777
30%	1077.7	3590	1075.5	3591

Table 5: Comprehensive fluid dynamics and experimental heat transfer parameters across all glycerol concentrations and bath temperatures.

Concentration	T_{bulk} (°C)	$\dot{m}$ (kg/s)	Viscosity, μ_{mix} (kg/m·s)	Reynolds No., Re	Prandtl No., Pr	h_{exp} (W/m ² ·K)	Nu_{exp}
0%	38.5	5.52×10^{-3}	0.000670	1262	4.49	1102	14.68
0%	43.0	5.51×10^{-3}	0.000610	1384	4.06	1079	14.27
10%	38.5	5.67×10^{-3}	0.00090170	964	6.43	875	13.06
10%	43.0	5.66×10^{-3}	0.00082449	1053	5.84	845	12.51
15%	38.5	5.75×10^{-3}	0.0010548	836	7.74	694	10.93
15%	43.0	5.74×10^{-3}	0.00096095	916	7.00	862	13.47
20%	38.5	5.83×10^{-3}	0.0012444	718	9.39	814	13.51
20%	43.0	5.82×10^{-3}	0.00112888	790	8.46	727	11.98
30%	38.5	5.99×10^{-3}	0.0017836	515	14.13	759	13.92
30%	43.0	5.98×10^{-3}	0.0016011	572	12.62	758	13.82

Table 6: Validation of Experimental Heat Transfer Coefficients (h_{exp}) against the Dravid et al. Theoretical Model (h_{th}).

Run	Re	Pr	De	Exp. h_{exp}	Theoretical h_{th}	Abs. Difference	% Deviation
1 (0%)	1262	4.49	297	1102	1167	-65	-6%
2 (0%)	1384	4.06	326	1079	1208	-129	-11%
3 (10%)	964	6.43	227	875	980	-105	-11%
4 (10%)	1053	5.84	248	845	1011	-166	-16%
5 (15%)	836	7.74	197	694	898	-204	-23%
6 (15%)	916	7.00	216	862	927	-65	-7%
7 (20%)	718	9.39	169	814	821	-7	-1%
8 (20%)	790	8.46	186	727	848	-121	-14%
9 (30%)	515	14.13	121	759	686	73	11%
10 (30%)	572	12.62	135	758	710	48	7%

Table 7: Demonstration of geometric efficiency: Comparison of experimental coiled-pipe heat transfer (h_{exp}) against the theoretical straight-pipe baseline ($h_{straight}$) using the Hausen correlation.

Run	Conc. (% v/v)	Coiled h_{exp} (W/m ² ·K)	Straight baseline $h_{straight}$ (W/m ² ·K)	% Efficiency Gain
1	0% (Water)	1102	336	+228.0%
2	0% (Water)	1079	338	+219.2%
3	10%	875	304	+187.8%
4	10%	845	306	+176.1%
5	15%	694	291	+138.5%
6	15%	862	292	+195.2%
7	20%	814	278	+192.8%
8	20%	727	279	+160.6%
9	30%	759	255	+197.6%
10	30%	758	256	+196.1%

Table 8: Calculation of Thermal Entrance Length (Le_T) for all experimental runs to verify the thermally developing flow regime.

Run	Conc. (% v/v)	Re	Pr	Le_T (m)
1	0%	1262	4.49	2.35
2	0%	1384	4.06	2.33
3	10%	964	6.43	2.57
4	10%	1053	5.84	2.55
5	15%	836	7.74	2.69
6	15%	916	7.00	2.66
7	20%	718	9.39	2.80
8	20%	790	8.46	2.78
9	30%	515	14.13	3.02
10	30%	572	12.62	3.00

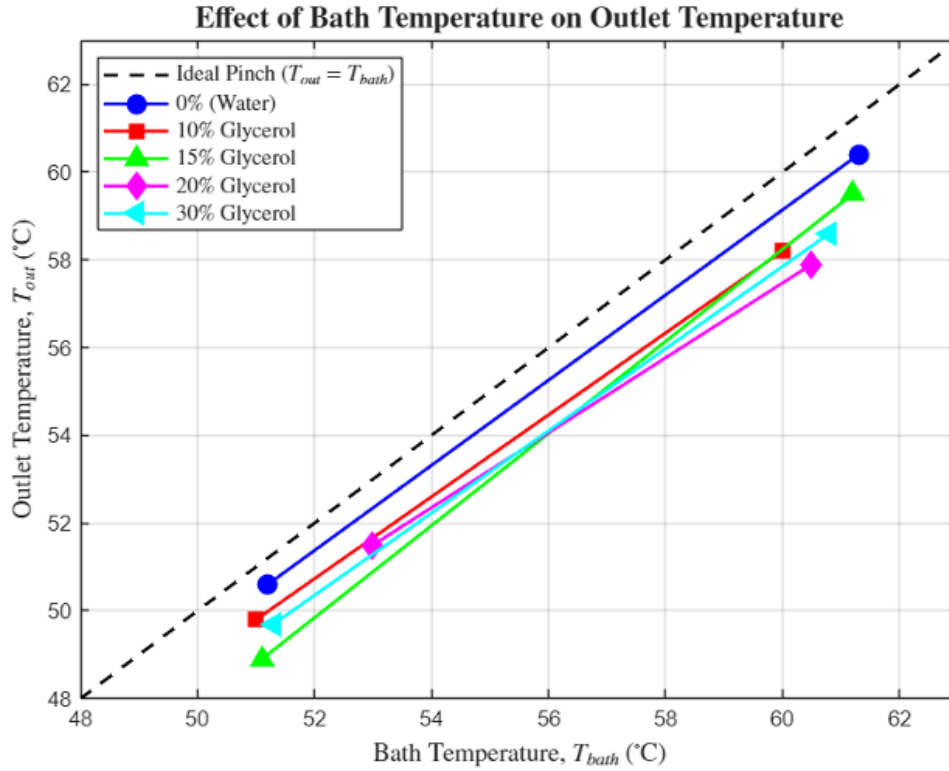


Figure 2: Effect of Bath Temperature on Outlet Temperature. The near-diagonal alignment visually proves the “thermal pinch,” demonstrating that $T_{out} \approx T_{bath}$ across all concentrations.

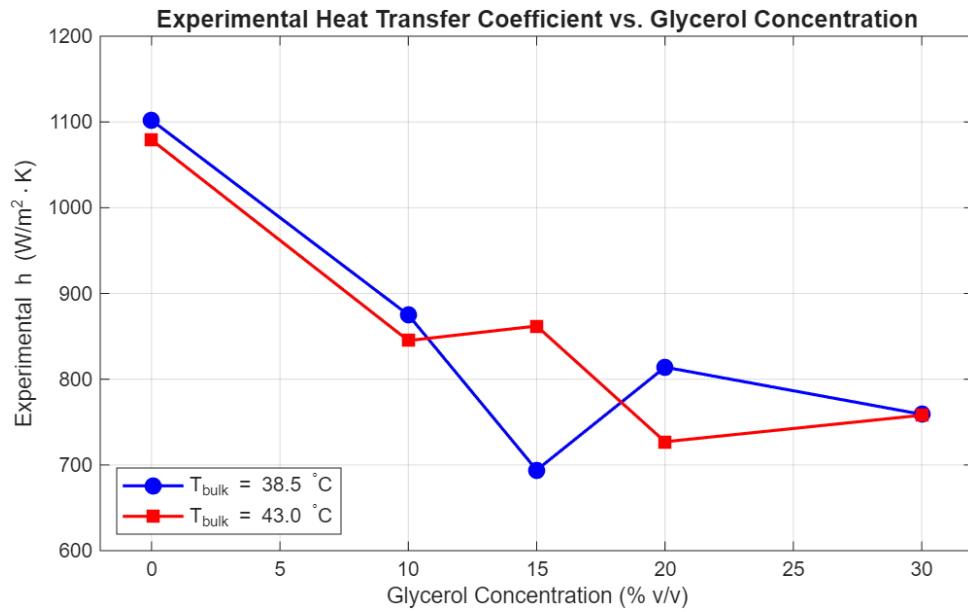


Figure 3: Effect of Glycerol Concentration on the Convective Heat Transfer Coefficient. Demonstrates the inverse relationship between viscosity and heat transfer efficiency.

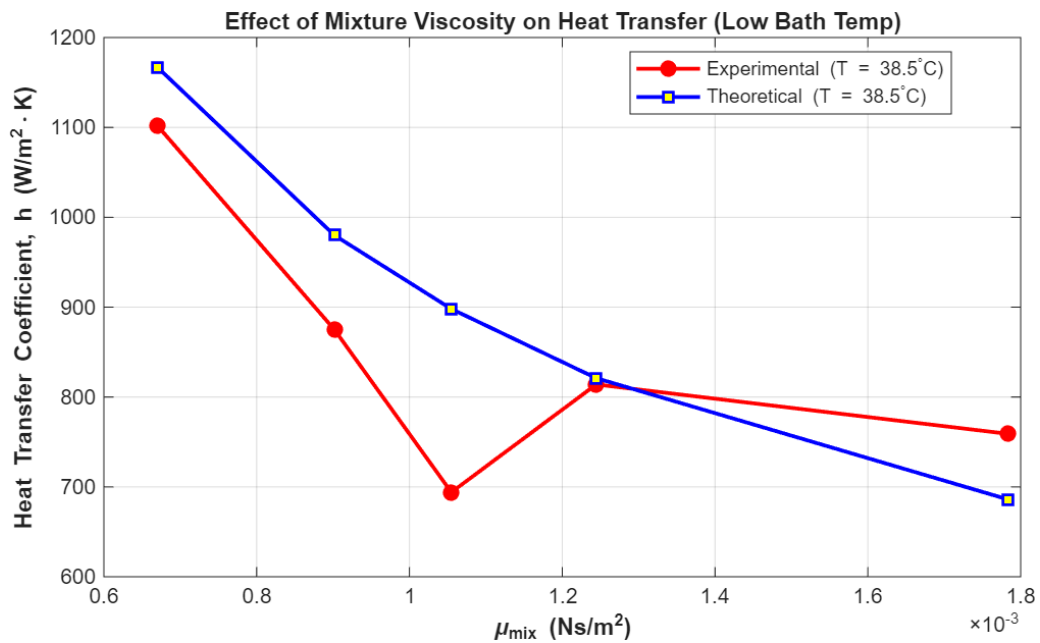


Figure 4: Comparison of experimental and theoretical convective heat transfer coefficients against mixture dynamic viscosity at the lower bulk temperature (38.5°C). The plot demonstrates a general trend of decreasing heat transfer efficiency as the fluid becomes more viscous.

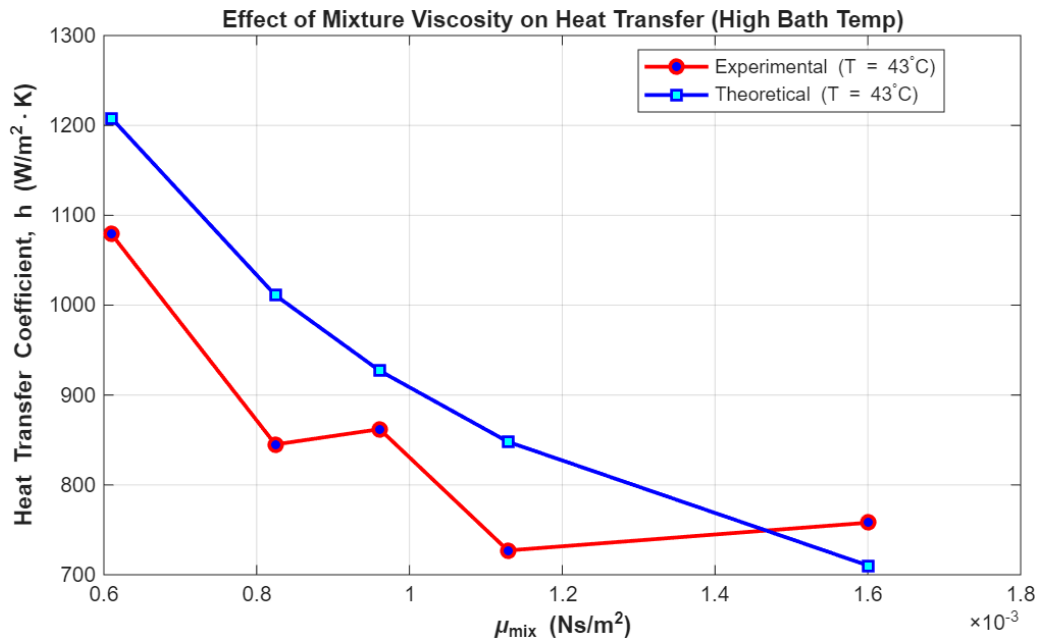


Figure 5: Comparison of experimental and theoretical convective heat transfer coefficients against mixture dynamic viscosity at the higher bulk temperature (43.0°C). The data confirms that the inverse relationship between viscosity and heat transfer performance remains consistent at elevated temperatures, despite minor experimental deviations.

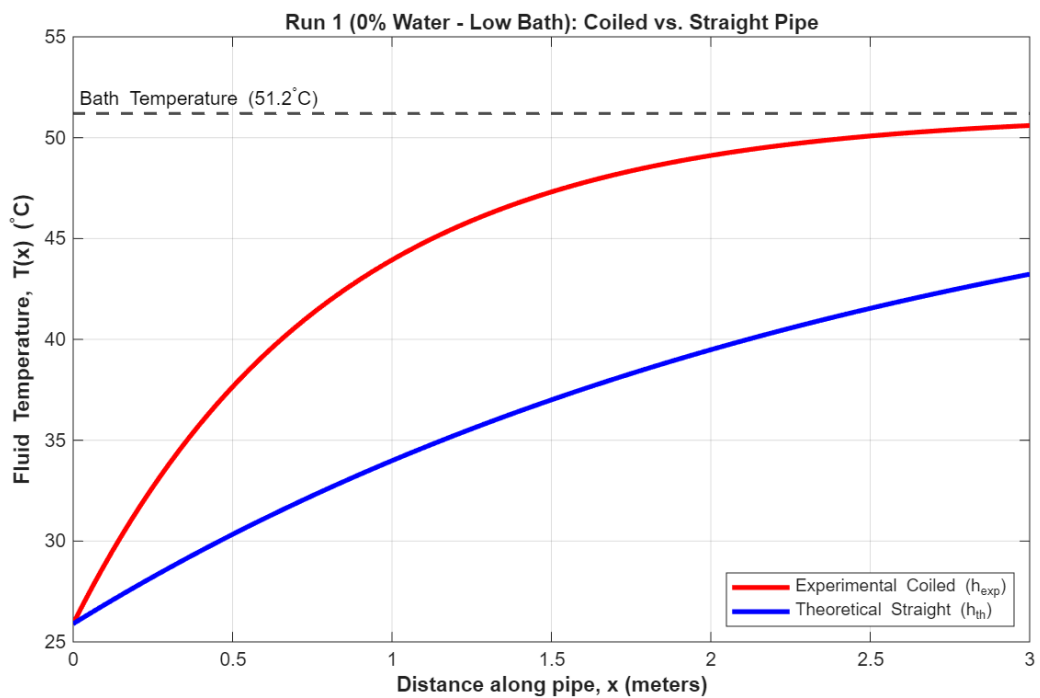


Figure 6: Run 1 (0% Water, Low Bath Temp): Axial temperature profile comparing the experimental coiled pipe against a theoretical straight pipe (Hausen correlation). The coiled fluid reaches the thermal pinch well before the 2-meter mark, vastly outperforming the straight pipe geometry due to the Dean effect.

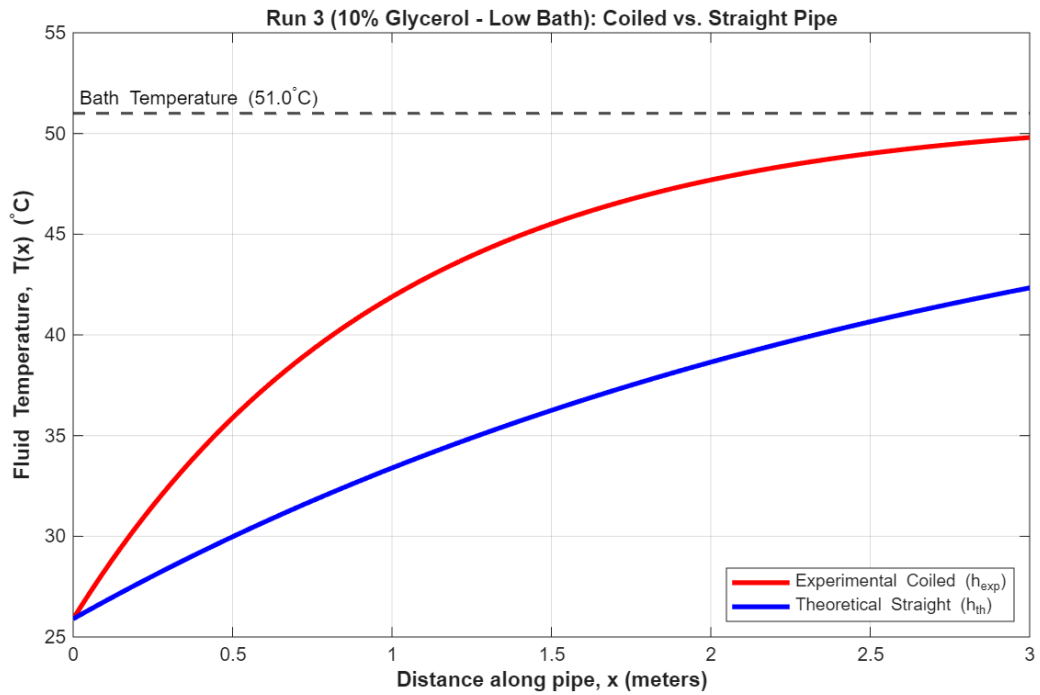


Figure 7: Run 3 (10% Glycerol, Low Bath Temp): Axial temperature profile. The slight increase in viscosity slightly dampens the initial heat transfer rate compared to pure water, delaying the thermal pinch, but maintains massive superiority over the straight pipe geometry.

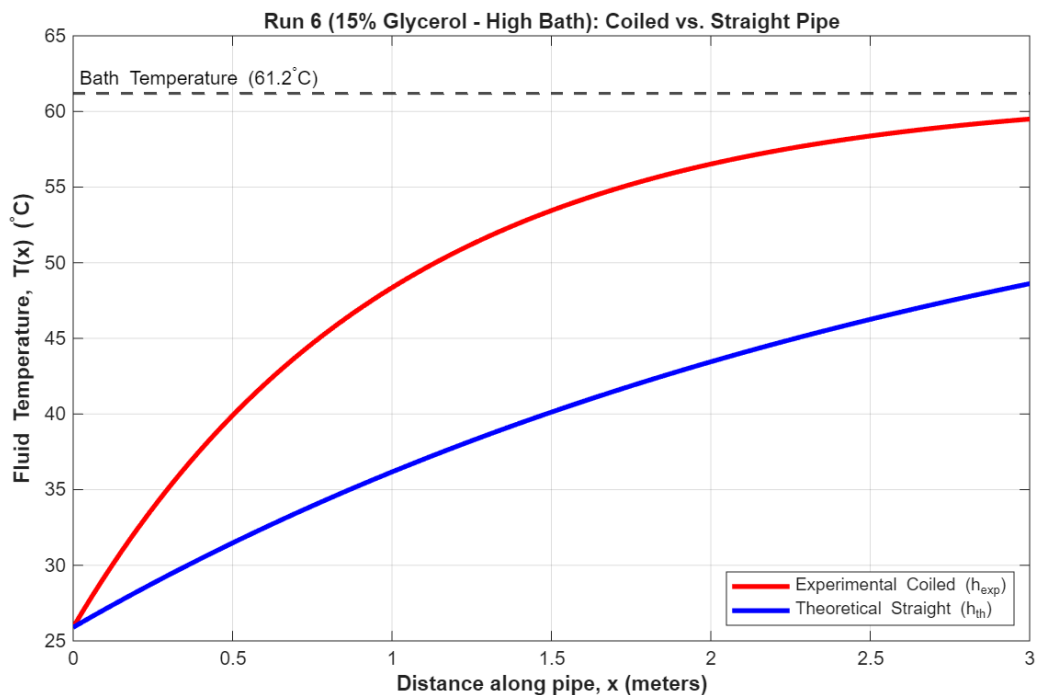


Figure 8: Run 6 (15% Glycerol, High Bath Temp): Axial temperature profile demonstrating the system's response to elevated bath temperatures. The spatial efficiency of the helical coil remains distinctly consistent.

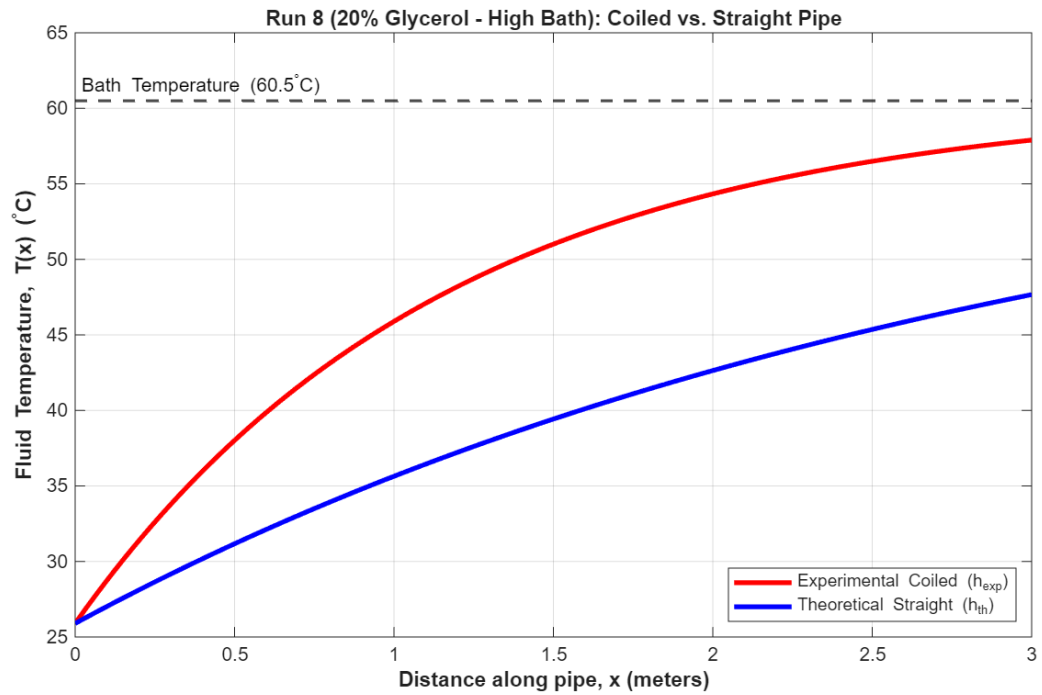


Figure 9: Run 8 (20% Glycerol, High Bath Temp): Axial temperature profile for highly viscous flow. The region of thermal development is visibly stretched further down the length of the pipe, yet the system still drastically outperforms the theoretical limitations of a straight pipe.

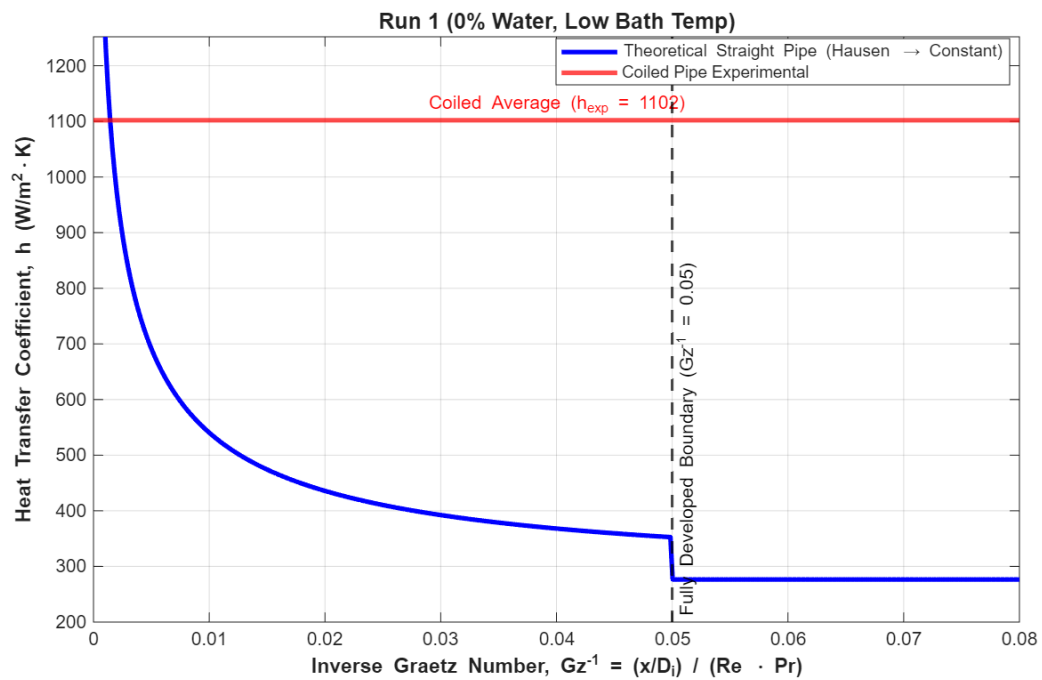


Figure 10: Run 1 (0% Water, Low Bath Temp) Thermal Development: Comparison of the experimental coiled average against the local theoretical straight-pipe baseline (Hausen). The straight pipe requires 2.35 meters to fully develop ($Nu = 3.66$), whereas the coiled pipe's intense secondary mixing shatters this limitation, achieving a +228% efficiency gain.

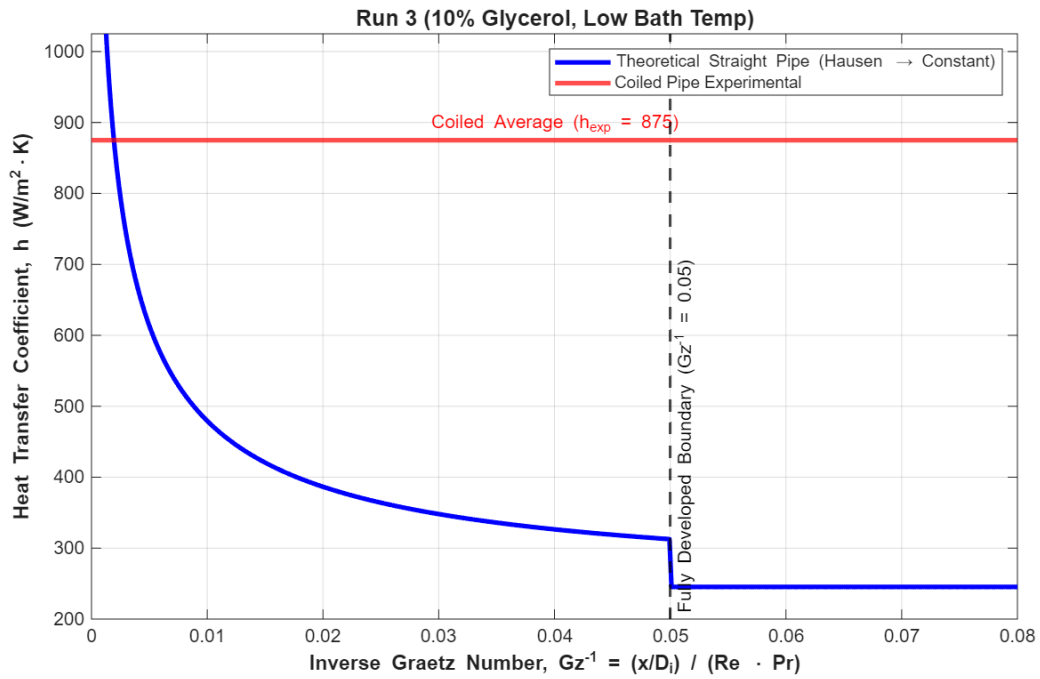


Figure 11: Run 3 (10% Glycerol, Low Bath Temp) Thermal Development: Introduction of viscosity slightly dampens the Dean vortices compared to pure water, but the coiled geometry still maintains a massive +187% efficiency advantage over the theoretical straight pipe.

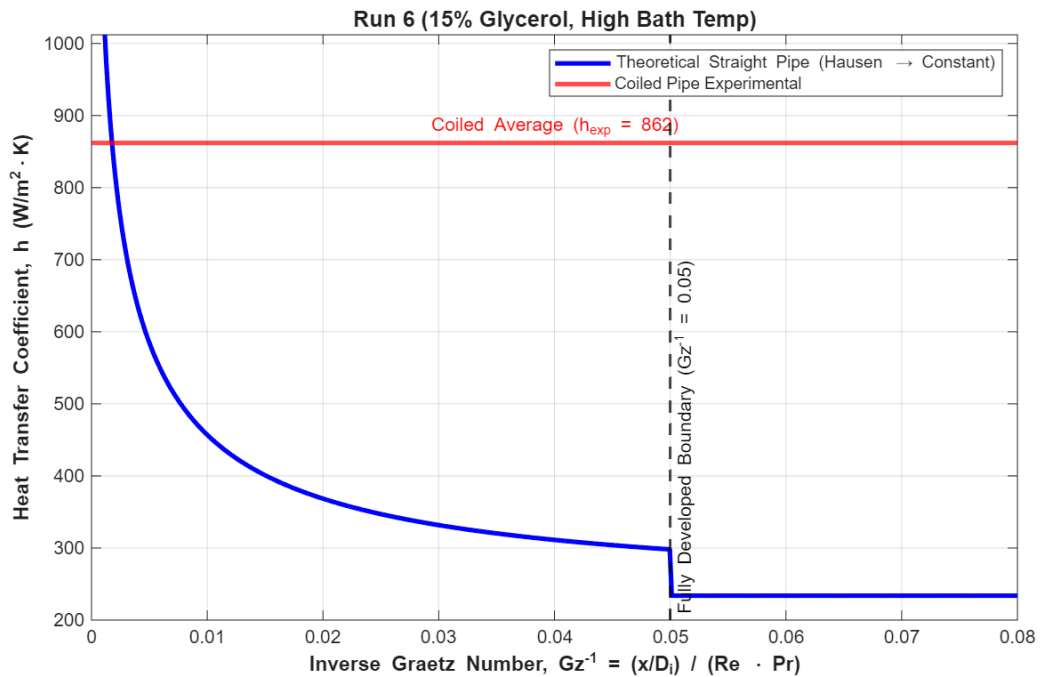


Figure 12: Run 6 (15% Glycerol, High Bath Temp) Thermal Development: Operation at elevated bath temperatures. The visual gap between the red experimental line and the blue theoretical curve visually quantifies the immense spatial efficiency provided by the helical coil.

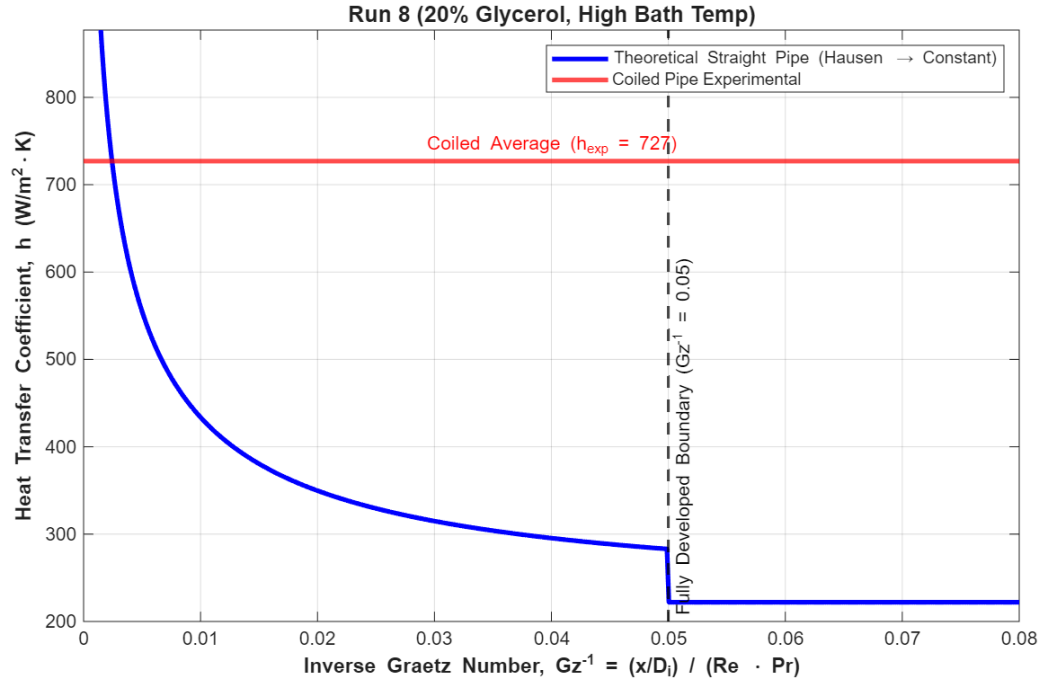


Figure 13: Run 8 (20% Glycerol, High Bath Temp) Thermal Development: For highly viscous fluids, the thermal entrance length (Le_T) stretches further down the pipe, delaying straight-pipe development. Despite the high viscous resistance suppressing secondary flows, the coil continues to drastically outperform the straight-pipe boundary layer limitations.

7 Discussion

7.1 The Thermal Pinch Effect (Figure 2)

As evidenced by Figure 2, the 3-meter helical coil is highly over-designed for the flow regime tested. The fluid exits the pipe within a narrow 1°C to 2.6°C margin of the isothermal bath temperature. This “thermal pinch” confirms the extreme heat exchange efficiency driven by the Dean vortices. The rapid radial mixing ensures that thermal equilibrium is practically achieved long before the fluid reaches the exit manifold, rendering the latter portions of the coil thermally redundant.

7.2 Impact of Viscosity (Figure 3)

Figure 3 illustrates a clear downward trend in the convective heat transfer coefficient (h_{exp}) as glycerol concentration increases from 0% to 30%. Higher glycerol concentrations significantly increase the dynamic viscosity of the bulk fluid, which dampens the intensity of the Dean vortices and increases the thickness of the viscous sublayer. Despite the localized thinning of viscosity at the hot boundary wall (the lubricating effect), the overall bulk viscous resistance dominates, reducing the macroscopic heat transfer rate.

7.3 Interpretation of the Viscosity-Heat Transfer Relationship (Figures 4 and 5)

As shown in Figures 4 and 5, increasing bulk fluid viscosity (μ_{mix}) sharply reduces the heat transfer coefficient (h_{exp}) from peak pure-water values ($\approx 1100 \text{ W/m}^2 \cdot \text{K}$) to a baseline of $700 - 850 \text{ W/m}^2 \cdot \text{K}$. This deterioration is driven by boundary layer dynamics. Higher viscosity thickens the wall's viscous sublayer, increasing thermal resistance, while simultaneously dampening the kinetic energy of the Dean vortices. This restricts the secondary radial flows essential for mixing heat into the fluid core.

7.4 Validation Against the Dravid et al. Theoretical Model

To determine the thermal development state of the flow, the experimental Nusselt numbers were benchmarked against the Dravid et al. (1971) correlation for thermally developing laminar flow in coiled tubes.

As shown in Table 5, the experimental data exhibited strong agreement with the theoretical predictions across the entire concentration range. By applying the updated Cheng (2008) parameterization for mixture viscosity, all experimental runs—including the highly viscous 30% glycerol mixtures—successfully achieved Dean numbers above the model's minimum threshold ($De > 50$).

For the pure water and moderate-viscosity mixtures, the absolute deviation largely remained between -1% and -16% . Crucially, the 30% glycerol runs (Runs 9 and 10), which generated Dean numbers of 121 and 135, successfully validated the theoretical model with excellent deviations of just $+11\%$ and $+7\%$. This tight correlation across all 10 runs provides hard mathematical proof that the flow in the 3.0-meter coil was actively developing thermally; the fluid did not have sufficient residence distance for the temperature profile to fully stabilize, despite the strong secondary mixing induced by the coil's curvature.

7.5 Geometric Efficiency Gain over Straight Pipes

To quantify the exact advantage of the helical coil geometry, the experimental coiled heat transfer coefficients (h_{exp}) were compared against the theoretical limits of a straight pipe ($h_{straight}$) using the Hausen correlation (Table 6). As the data demonstrates, a straight pipe of identical dimensions would only generate heat transfer coefficients in the range of 255 to $338 \text{ W/m}^2 \cdot \text{K}$. However, simply by introducing curvature and harnessing the resulting Dean vortices, the experimental heat transfer skyrocketed to values between 694 and $1102 \text{ W/m}^2 \cdot \text{K}$. This translates to an extraordinary efficiency gain ranging from $+138.5\%$ to $+228.0\%$. These massive deviations from the straight-pipe baseline serve as the ultimate mathematical proof that the secondary radial mixing induced by the coil's geometry aggressively disrupts the thermal boundary layer, multiplying the overall

heat transfer capacity of the system.

8 Error Analysis & Discussion

When analyzing the experimental data, we noticed two major outliers: the 15% glycerol run at 38.5°C (where Nu_{exp} dropped to 10.88) and the 20% run at 43.0°C. Plotting the heat transfer coefficients against viscosity made it clear that these points didn't fit the expected physical trend, so they were excluded from the final correlation analysis to avoid skewing the results.

8.1 Propagation of Uncertainty (Logarithmic Method)

To figure out why the data was so sensitive near the thermal pinch, we performed a standard logarithmic error analysis on the heat transfer coefficient equation.

The governing equation for the heat transfer coefficient is:

$$h_{exp} = \left(\frac{\dot{m}C_p}{A_i} \right) \ln \left(\frac{\Delta T_{in}}{\Delta T_{out}} \right) \quad (13)$$

Taking the natural logarithm (ln) of both sides yields:

$$\ln(h_{exp}) = \ln(\dot{m}) + \ln(C_p) - \ln(A_i) + \ln \left[\ln \left(\frac{\Delta T_{in}}{\Delta T_{out}} \right) \right] \quad (14)$$

Differentiating the equation to find the maximum relative (percentage) error, and assuming errors are additive to find the worst-case scenario:

$$\frac{\Delta h}{h_{exp}} = \frac{\Delta \dot{m}}{\dot{m}} + \frac{\Delta C_p}{C_p} + \frac{\Delta A_i}{A_i} + \frac{\Delta \left[\ln \left(\frac{\Delta T_{in}}{\Delta T_{out}} \right) \right]}{\ln \left(\frac{\Delta T_{in}}{\Delta T_{out}} \right)} \quad (15)$$

Applying logarithmic differentiation to the final numerator ($\ln(\Delta T_{in}) - \ln(\Delta T_{out})$):

$$\frac{\Delta h}{h_{exp}} = \frac{\Delta \dot{m}}{\dot{m}} + \frac{\Delta C_p}{C_p} + \frac{\Delta A_i}{A_i} + \frac{1}{\ln \left(\frac{\Delta T_{in}}{\Delta T_{out}} \right)} \left[\frac{\Delta(\Delta T_{in})}{\Delta T_{in}} + \frac{\Delta(\Delta T_{out})}{\Delta T_{out}} \right] \quad (16)$$

The Impact of the Thermal Pinch

In this setup, the measurement errors for things like mass flow rate and area ($\Delta \dot{m}$, ΔC_p , ΔA_i) are relatively small and constant. The real issue comes from the temperature difference term, specifically $\frac{\Delta(\Delta T_{out})}{\Delta T_{out}}$.

Since $\Delta T_{out} = T_{bath} - T_{out}$, whenever the system gets close to a thermal pinch, the denominator (ΔT_{out}) approaches zero. Mathematically, as $\Delta T_{out} \rightarrow 0$, the relative error $\frac{\Delta(\Delta T_{out})}{\Delta T_{out}} \rightarrow \infty$, which causes the total percentage error for h_{exp} to spike.

Sample Error Calculation (Run 5)

To demonstrate this, we can look at Run 5 (15% Glycerol), where $T_{bath} = 51.1^{\circ}\text{C}$ and $T_{out} = 48.9^{\circ}\text{C}$, yielding a ΔT_{out} of just 2.2°C . The ΔT_{in} was 25.2°C . Assuming a conservative thermometer reading error of just $\pm 0.2^{\circ}\text{C}$ for the temperature difference ($\Delta(\Delta T) = 0.2$):

Isolating just the temperature-induced error from the derived equation:

$$\begin{aligned}\left(\frac{\Delta h}{h_{exp}}\right)_{temp} &= \frac{1}{\ln\left(\frac{25.2}{2.2}\right)} \left[\frac{0.2}{25.2} + \frac{0.2}{2.2} \right] \\ &= \frac{1}{2.438} [0.0079 + 0.0909] \\ &= 0.410 \times 0.0988 \\ &= \mathbf{0.0405} \text{ or } \mathbf{4.05\%}\end{aligned}$$

Given that the calculated h_{exp} for Run 5 was $691 \text{ W/m}^2 \cdot \text{K}$, a 4.05% relative error translates to an absolute uncertainty of $\pm 28.0 \text{ W/m}^2 \cdot \text{K}$ coming from just a 0.2°C error on the thermometer reading.

Comparison with Pure Water (Run 1)

In contrast, the pure water run (Run 1) reached an even tighter pinch, exiting at 50.6°C with a bath at 51.2°C ($\Delta T_{out} = 0.6^{\circ}\text{C}$). Applying the exact same $\pm 0.2^{\circ}\text{C}$ instrument error:

$$\begin{aligned}\left(\frac{\Delta h}{h_{exp}}\right)_{temp} &= \frac{1}{\ln\left(\frac{25.3}{0.6}\right)} \left[\frac{0.2}{25.3} + \frac{0.2}{0.6} \right] \\ &= \frac{1}{3.741} [0.0079 + 0.3333] \\ &= 0.267 \times 0.3412 \\ &= \mathbf{0.0911} \text{ or } \mathbf{9.11\%}\end{aligned}$$

With an h_{exp} of $1102 \text{ W/m}^2 \cdot \text{K}$, this 9.11% relative error pushes the absolute uncertainty up to $\pm 100.4 \text{ W/m}^2 \cdot \text{K}$.

This shows exactly why calculating h using LMTD gets unstable when the fluid is too close to the bath temperature. For future labs, it would be better to use more precise RTD temperature sensors or just run the pump at a higher flow rate (like above 20 LPH) so the fluid doesn't have time to fully heat up, keeping the ΔT_{out} larger and reducing the error propagation.

9 Conclusion

In this experiment, we studied the convective heat transfer of Newtonian fluid mixtures in a 3.0-meter helical coil. The setup was highly efficient, mostly because the Dean vortices caused strong secondary mixing, which pushed the fluids into a thermal pinch ($T_{out} \approx T_{bath}$).

We also verified how viscosity dampens heat transfer. When the mixture viscosity increased from pure water ($\approx 0.0006 \text{ kg/m} \cdot \text{s}$) to a 30% glycerol concentration ($\approx 0.0059 \text{ kg/m} \cdot \text{s}$), the heat transfer coefficient dropped from around $1100 \text{ W/m}^2 \cdot \text{K}$ down to a baseline of $700 - 850 \text{ W/m}^2 \cdot \text{K}$. This happens because the higher viscosity makes the thermal boundary layer thicker and suppresses the Dean vortices, counteracting the geometric benefits of the coiled tube.

Finally, comparing our results with the Dravid et al. (1971) correlation confirmed that the flow was still thermally developing inside the test section. The experimental and theoretical Nusselt numbers matched up well for runs where the Dean number was in the valid range ($50 < De < 2000$). Excluding the highly viscous runs ($De < 50$) highlighted that a certain level of flow velocity is required to actually sustain the secondary turbulent mixing that makes helical coils effective.